AI_Business_Research_Copilot

October 6, 2025

AI Business Research Copilot (RAG + Local Ollama)

Author: Akshay Bhujbal

Project Type: Generative AI / NLP / Retrieval-Augmented Generation (RAG) / Local LLM Portfolio Project

Project Overview

This project demonstrates a **Business Research Copilot system** using:

- **LangChain + Ollama (Local LLM)**
- **ChromaDB (Vector database for semantic search)**
- **WebBaseLoader (Load website content dynamically)**

The project allows users to:

1. Load multiple trusted business/finance websites.
2. Split website content into chunks for efficient processing.
3. Create embeddings and store them in a vector database.
4. Query the LLM to get answers with source citations.
5. Build a fully offline, local AI-powered research assistant.

Websites used (trusted sources):

- MoneyControl
- Investing.com
- Economic Times
- Business Standard

Tech Stack:

- Python
- LangChain
- ChromaDB (Vector DB)
- Ollama Llama3.2:1b (local model)
- Sentence Transformers (Embeddings)
- Streamlit (optional for deployment)

1 Install Required Libraries

- `langchain` → **Orchestrates LLM + vector DB + retrieval.**
- `chromadb` → **Stores embeddings for retrieval.**
- `sentence-transformers` → **Converts text into embeddings.**
- `beautifulsoup4` + `lxml` + `requests` → **Scrape web pages.**

[1]:
```
# Install all required packages
!pip install langchain chromadb sentence-transformers beautifulsoup4 requests␣
↪lxml
```

```
Requirement already satisfied: langchain in c:\users\lenovo\anaconda3\lib\site-
packages (0.3.27)
Requirement already satisfied: chromadb in c:\users\lenovo\anaconda3\lib\site-
packages (1.1.0)
Requirement already satisfied: sentence-transformers in
c:\users\lenovo\anaconda3\lib\site-packages (5.1.1)
Requirement already satisfied: beautifulsoup4 in
c:\users\lenovo\anaconda3\lib\site-packages (4.12.3)
Requirement already satisfied: requests in c:\users\lenovo\anaconda3\lib\site-
packages (2.32.5)
Requirement already satisfied: lxml in c:\users\lenovo\anaconda3\lib\site-
packages (5.3.0)
Requirement already satisfied: langchain-core<1.0.0,>=0.3.72 in
c:\users\lenovo\anaconda3\lib\site-packages (from langchain) (0.3.76)
Requirement already satisfied: langchain-text-splitters<1.0.0,>=0.3.9 in
c:\users\lenovo\anaconda3\lib\site-packages (from langchain) (0.3.11)
Requirement already satisfied: langsmith>=0.1.17 in
c:\users\lenovo\anaconda3\lib\site-packages (from langchain) (0.4.29)
Requirement already satisfied: pydantic<3.0.0,>=2.7.4 in
c:\users\lenovo\anaconda3\lib\site-packages (from langchain) (2.10.3)
Requirement already satisfied: SQLAlchemy<3,>=1.4 in
c:\users\lenovo\anaconda3\lib\site-packages (from langchain) (2.0.39)
Requirement already satisfied: PyYAML>=5.3 in
c:\users\lenovo\anaconda3\lib\site-packages (from langchain) (6.0.2)
Requirement already satisfied: charset_normalizer<4,>=2 in
c:\users\lenovo\anaconda3\lib\site-packages (from requests) (3.3.2)
Requirement already satisfied: idna<4,>=2.5 in
c:\users\lenovo\anaconda3\lib\site-packages (from requests) (3.7)
Requirement already satisfied: urllib3<3,>=1.21.1 in
c:\users\lenovo\anaconda3\lib\site-packages (from requests) (2.3.0)
Requirement already satisfied: certifi>=2017.4.17 in
c:\users\lenovo\anaconda3\lib\site-packages (from requests) (2025.4.26)
Requirement already satisfied: tenacity!=8.4.0,<10.0.0,>=8.1.0 in
c:\users\lenovo\anaconda3\lib\site-packages (from langchain-
core<1.0.0,>=0.3.72->langchain) (9.0.0)
Requirement already satisfied: jsonpatch<2.0,>=1.33 in
c:\users\lenovo\anaconda3\lib\site-packages (from langchain-
```

```
core<1.0.0,>=0.3.72->langchain) (1.33)
Requirement already satisfied: typing-extensions>=4.7 in
c:\users\lenovo\anaconda3\lib\site-packages (from langchain-
core<1.0.0,>=0.3.72->langchain) (4.12.2)
Requirement already satisfied: packaging>=23.2 in
c:\users\lenovo\anaconda3\lib\site-packages (from langchain-
core<1.0.0,>=0.3.72->langchain) (24.2)
Requirement already satisfied: jsonpointer>=1.9 in
c:\users\lenovo\anaconda3\lib\site-packages (from
jsonpatch<2.0,>=1.33->langchain-core<1.0.0,>=0.3.72->langchain) (2.1)
Requirement already satisfied: annotated-types>=0.6.0 in
c:\users\lenovo\anaconda3\lib\site-packages (from
pydantic<3.0.0,>=2.7.4->langchain) (0.6.0)
Requirement already satisfied: pydantic-core==2.27.1 in
c:\users\lenovo\anaconda3\lib\site-packages (from
pydantic<3.0.0,>=2.7.4->langchain) (2.27.1)
Requirement already satisfied: greenlet!=0.4.17 in
c:\users\lenovo\anaconda3\lib\site-packages (from SQLAlchemy<3,>=1.4->langchain)
(3.1.1)
Requirement already satisfied: build>=1.0.3 in
c:\users\lenovo\anaconda3\lib\site-packages (from chromadb) (1.3.0)
Requirement already satisfied: pybase64>=1.4.1 in
c:\users\lenovo\anaconda3\lib\site-packages (from chromadb) (1.4.2)
Requirement already satisfied: uvicorn>=0.18.3 in
c:\users\lenovo\anaconda3\lib\site-packages (from
uvicorn[standard]>=0.18.3->chromadb) (0.34.3)
Requirement already satisfied: numpy>=1.22.5 in
c:\users\lenovo\anaconda3\lib\site-packages (from chromadb) (2.3.3)
Requirement already satisfied: posthog<6.0.0,>=2.4.0 in
c:\users\lenovo\anaconda3\lib\site-packages (from chromadb) (5.4.0)
Requirement already satisfied: onnxruntime>=1.14.1 in
c:\users\lenovo\anaconda3\lib\site-packages (from chromadb) (1.22.1)
Requirement already satisfied: opentelemetry-api>=1.2.0 in
c:\users\lenovo\anaconda3\lib\site-packages (from chromadb) (1.37.0)
Requirement already satisfied: opentelemetry-exporter-otlp-proto-grpc>=1.2.0 in
c:\users\lenovo\anaconda3\lib\site-packages (from chromadb) (1.37.0)
Requirement already satisfied: opentelemetry-sdk>=1.2.0 in
c:\users\lenovo\anaconda3\lib\site-packages (from chromadb) (1.37.0)
Requirement already satisfied: tokenizers>=0.13.2 in
c:\users\lenovo\anaconda3\lib\site-packages (from chromadb) (0.22.1)
Requirement already satisfied: pypika>=0.48.9 in
c:\users\lenovo\anaconda3\lib\site-packages (from chromadb) (0.48.9)
Requirement already satisfied: tqdm>=4.65.0 in
c:\users\lenovo\anaconda3\lib\site-packages (from chromadb) (4.67.1)
Requirement already satisfied: overrides>=7.3.1 in
c:\users\lenovo\anaconda3\lib\site-packages (from chromadb) (7.4.0)
Requirement already satisfied: importlib-resources in
c:\users\lenovo\anaconda3\lib\site-packages (from chromadb) (6.5.2)
```

```
Requirement already satisfied: grpcio>=1.58.0 in
c:\users\lenovo\anaconda3\lib\site-packages (from chromadb) (1.73.1)
Requirement already satisfied: bcrypt>=4.0.1 in
c:\users\lenovo\anaconda3\lib\site-packages (from chromadb) (4.3.0)
Requirement already satisfied: typer>=0.9.0 in
c:\users\lenovo\anaconda3\lib\site-packages (from chromadb) (0.16.0)
Requirement already satisfied: kubernetes>=28.1.0 in
c:\users\lenovo\anaconda3\lib\site-packages (from chromadb) (33.1.0)
Requirement already satisfied: mmh3>=4.0.1 in
c:\users\lenovo\anaconda3\lib\site-packages (from chromadb) (5.2.0)
Requirement already satisfied: orjson>=3.9.12 in
c:\users\lenovo\anaconda3\lib\site-packages (from chromadb) (3.10.18)
Requirement already satisfied: httpx>=0.27.0 in
c:\users\lenovo\anaconda3\lib\site-packages (from chromadb) (0.28.1)
Requirement already satisfied: rich>=10.11.0 in
c:\users\lenovo\anaconda3\lib\site-packages (from chromadb) (13.9.4)
Requirement already satisfied: jsonschema>=4.19.0 in
c:\users\lenovo\anaconda3\lib\site-packages (from chromadb) (4.23.0)
Requirement already satisfied: six>=1.5 in c:\users\lenovo\anaconda3\lib\site-
packages (from posthog<6.0.0,>=2.4.0->chromadb) (1.17.0)
Requirement already satisfied: python-dateutil>=2.2 in
c:\users\lenovo\anaconda3\lib\site-packages (from
posthog<6.0.0,>=2.4.0->chromadb) (2.9.0.post0)
Requirement already satisfied: backoff>=1.10.0 in
c:\users\lenovo\anaconda3\lib\site-packages (from
posthog<6.0.0,>=2.4.0->chromadb) (1.10.0)
Requirement already satisfied: distro>=1.5.0 in
c:\users\lenovo\anaconda3\lib\site-packages (from
posthog<6.0.0,>=2.4.0->chromadb) (1.9.0)
Requirement already satisfied: transformers<5.0.0,>=4.41.0 in
c:\users\lenovo\anaconda3\lib\site-packages (from sentence-transformers)
(4.56.2)
Requirement already satisfied: torch>=1.11.0 in
c:\users\lenovo\anaconda3\lib\site-packages (from sentence-transformers) (2.8.0)
Requirement already satisfied: scikit-learn in
c:\users\lenovo\anaconda3\lib\site-packages (from sentence-transformers) (1.7.0)
Requirement already satisfied: scipy in c:\users\lenovo\anaconda3\lib\site-
packages (from sentence-transformers) (1.15.3)
Requirement already satisfied: huggingface-hub>=0.20.0 in
c:\users\lenovo\anaconda3\lib\site-packages (from sentence-transformers)
(0.35.3)
Requirement already satisfied: Pillow in c:\users\lenovo\anaconda3\lib\site-
packages (from sentence-transformers) (11.1.0)
Requirement already satisfied: filelock in c:\users\lenovo\anaconda3\lib\site-
packages (from transformers<5.0.0,>=4.41.0->sentence-transformers) (3.17.0)
Requirement already satisfied: regex!=2019.12.17 in
c:\users\lenovo\anaconda3\lib\site-packages (from
transformers<5.0.0,>=4.41.0->sentence-transformers) (2024.11.6)
```

```
Requirement already satisfied: safetensors>=0.4.3 in
c:\users\lenovo\anaconda3\lib\site-packages (from
transformers<5.0.0,>=4.41.0->sentence-transformers) (0.5.3)
Requirement already satisfied: fsspec>=2023.5.0 in
c:\users\lenovo\anaconda3\lib\site-packages (from huggingface-
hub>=0.20.0->sentence-transformers) (2025.3.2)
Requirement already satisfied: soupsieve>1.2 in
c:\users\lenovo\anaconda3\lib\site-packages (from beautifulsoup4) (2.5)
Requirement already satisfied: pyproject_hooks in
c:\users\lenovo\anaconda3\lib\site-packages (from build>=1.0.3->chromadb)
(1.2.0)
Requirement already satisfied: colorama in c:\users\lenovo\anaconda3\lib\site-
packages (from build>=1.0.3->chromadb) (0.4.6)
Requirement already satisfied: anyio in c:\users\lenovo\anaconda3\lib\site-
packages (from httpx>=0.27.0->chromadb) (4.11.0)
Requirement already satisfied: httpcore==1.* in
c:\users\lenovo\anaconda3\lib\site-packages (from httpx>=0.27.0->chromadb)
(1.0.9)
Requirement already satisfied: h11>=0.16 in c:\users\lenovo\anaconda3\lib\site-
packages (from httpcore==1.*->httpx>=0.27.0->chromadb) (0.16.0)
Requirement already satisfied: attrs>=22.2.0 in
c:\users\lenovo\anaconda3\lib\site-packages (from jsonschema>=4.19.0->chromadb)
(24.3.0)
Requirement already satisfied: jsonschema-specifications>=2023.03.6 in
c:\users\lenovo\anaconda3\lib\site-packages (from jsonschema>=4.19.0->chromadb)
(2023.7.1)
Requirement already satisfied: referencing>=0.28.4 in
c:\users\lenovo\anaconda3\lib\site-packages (from jsonschema>=4.19.0->chromadb)
(0.30.2)
Requirement already satisfied: rpds-py>=0.7.1 in
c:\users\lenovo\anaconda3\lib\site-packages (from jsonschema>=4.19.0->chromadb)
(0.22.3)
Requirement already satisfied: google-auth>=1.0.1 in
c:\users\lenovo\anaconda3\lib\site-packages (from kubernetes>=28.1.0->chromadb)
(2.40.3)
Requirement already satisfied: websocket-
client!=0.40.0,!=0.41.*,!=0.42.*,>=0.32.0 in c:\users\lenovo\anaconda3\lib\site-
packages (from kubernetes>=28.1.0->chromadb) (1.8.0)
Requirement already satisfied: requests-oauthlib in
c:\users\lenovo\anaconda3\lib\site-packages (from kubernetes>=28.1.0->chromadb)
(2.0.0)
Requirement already satisfied: oauthlib>=3.2.2 in
c:\users\lenovo\anaconda3\lib\site-packages (from kubernetes>=28.1.0->chromadb)
(3.3.1)
Requirement already satisfied: durationpy>=0.7 in
c:\users\lenovo\anaconda3\lib\site-packages (from kubernetes>=28.1.0->chromadb)
(0.10)
Requirement already satisfied: cachetools<6.0,>=2.0.0 in
```

```
c:\users\lenovo\anaconda3\lib\site-packages (from google-
auth>=1.0.1->kubernetes>=28.1.0->chromadb) (5.5.1)
Requirement already satisfied: pyasn1-modules>=0.2.1 in
c:\users\lenovo\anaconda3\lib\site-packages (from google-
auth>=1.0.1->kubernetes>=28.1.0->chromadb) (0.2.8)
Requirement already satisfied: rsa<5,>=3.1.4 in
c:\users\lenovo\anaconda3\lib\site-packages (from google-
auth>=1.0.1->kubernetes>=28.1.0->chromadb) (4.9.1)
Requirement already satisfied: pyasn1>=0.1.3 in
c:\users\lenovo\anaconda3\lib\site-packages (from rsa<5,>=3.1.4->google-
auth>=1.0.1->kubernetes>=28.1.0->chromadb) (0.4.8)
Requirement already satisfied: requests-toolbelt>=1.0.0 in
c:\users\lenovo\anaconda3\lib\site-packages (from langsmith>=0.1.17->langchain)
(1.0.0)
Requirement already satisfied: zstandard>=0.23.0 in
c:\users\lenovo\anaconda3\lib\site-packages (from langsmith>=0.1.17->langchain)
(0.23.0)
Requirement already satisfied: coloredlogs in
c:\users\lenovo\anaconda3\lib\site-packages (from onnxruntime>=1.14.1->chromadb)
(15.0.1)
Requirement already satisfied: flatbuffers in
c:\users\lenovo\anaconda3\lib\site-packages (from onnxruntime>=1.14.1->chromadb)
(25.2.10)
Requirement already satisfied: protobuf in c:\users\lenovo\anaconda3\lib\site-
packages (from onnxruntime>=1.14.1->chromadb) (5.29.3)
Requirement already satisfied: sympy in c:\users\lenovo\anaconda3\lib\site-
packages (from onnxruntime>=1.14.1->chromadb) (1.13.3)
Requirement already satisfied: importlib-metadata<8.8.0,>=6.0 in
c:\users\lenovo\anaconda3\lib\site-packages (from opentelemetry-
api>=1.2.0->chromadb) (8.5.0)
Requirement already satisfied: zipp>=3.20 in c:\users\lenovo\anaconda3\lib\site-
packages (from importlib-metadata<8.8.0,>=6.0->opentelemetry-
api>=1.2.0->chromadb) (3.21.0)
Requirement already satisfied: googleapis-common-protos~=1.57 in
c:\users\lenovo\anaconda3\lib\site-packages (from opentelemetry-exporter-otlp-
proto-grpc>=1.2.0->chromadb) (1.70.0)
Requirement already satisfied: opentelemetry-exporter-otlp-proto-common==1.37.0
in c:\users\lenovo\anaconda3\lib\site-packages (from opentelemetry-exporter-
otlp-proto-grpc>=1.2.0->chromadb) (1.37.0)
Requirement already satisfied: opentelemetry-proto==1.37.0 in
c:\users\lenovo\anaconda3\lib\site-packages (from opentelemetry-exporter-otlp-
proto-grpc>=1.2.0->chromadb) (1.37.0)
Requirement already satisfied: opentelemetry-semantic-conventions==0.58b0 in
c:\users\lenovo\anaconda3\lib\site-packages (from opentelemetry-
sdk>=1.2.0->chromadb) (0.58b0)
Requirement already satisfied: markdown-it-py>=2.2.0 in
c:\users\lenovo\anaconda3\lib\site-packages (from rich>=10.11.0->chromadb)
(2.2.0)
```

```
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in
c:\users\lenovo\anaconda3\lib\site-packages (from rich>=10.11.0->chromadb)
(2.19.1)
Requirement already satisfied: mdurl~=0.1 in c:\users\lenovo\anaconda3\lib\site-
packages (from markdown-it-py>=2.2.0->rich>=10.11.0->chromadb) (0.1.0)
Requirement already satisfied: networkx in c:\users\lenovo\anaconda3\lib\site-
packages (from torch>=1.11.0->sentence-transformers) (3.4.2)
Requirement already satisfied: jinja2 in c:\users\lenovo\anaconda3\lib\site-
packages (from torch>=1.11.0->sentence-transformers) (3.1.6)
Requirement already satisfied: setuptools in c:\users\lenovo\anaconda3\lib\site-
packages (from torch>=1.11.0->sentence-transformers) (72.1.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in
c:\users\lenovo\anaconda3\lib\site-packages (from
sympy->onnxruntime>=1.14.1->chromadb) (1.3.0)
Requirement already satisfied: click>=8.0.0 in
c:\users\lenovo\anaconda3\lib\site-packages (from typer>=0.9.0->chromadb)
(8.1.8)
Requirement already satisfied: shellingham>=1.3.0 in
c:\users\lenovo\anaconda3\lib\site-packages (from typer>=0.9.0->chromadb)
(1.5.0)
Requirement already satisfied: httptools>=0.6.3 in
c:\users\lenovo\anaconda3\lib\site-packages (from
uvicorn[standard]>=0.18.3->chromadb) (0.6.4)
Requirement already satisfied: python-dotenv>=0.13 in
c:\users\lenovo\anaconda3\lib\site-packages (from
uvicorn[standard]>=0.18.3->chromadb) (1.1.0)
Requirement already satisfied: watchfiles>=0.13 in
c:\users\lenovo\anaconda3\lib\site-packages (from
uvicorn[standard]>=0.18.3->chromadb) (1.1.0)
Requirement already satisfied: websockets>=10.4 in
c:\users\lenovo\anaconda3\lib\site-packages (from
uvicorn[standard]>=0.18.3->chromadb) (15.0.1)
Requirement already satisfied: sniffio>=1.1 in
c:\users\lenovo\anaconda3\lib\site-packages (from
anyio->httpx>=0.27.0->chromadb) (1.3.0)
Requirement already satisfied: humanfriendly>=9.1 in
c:\users\lenovo\anaconda3\lib\site-packages (from
coloredlogs->onnxruntime>=1.14.1->chromadb) (10.0)
Requirement already satisfied: pyreadline3 in
c:\users\lenovo\anaconda3\lib\site-packages (from
humanfriendly>=9.1->coloredlogs->onnxruntime>=1.14.1->chromadb) (3.5.4)
Requirement already satisfied: MarkupSafe>=2.0 in
c:\users\lenovo\anaconda3\lib\site-packages (from
jinja2->torch>=1.11.0->sentence-transformers) (3.0.2)
Requirement already satisfied: joblib>=1.2.0 in
c:\users\lenovo\anaconda3\lib\site-packages (from scikit-learn->sentence-
transformers) (1.4.2)
Requirement already satisfied: threadpoolctl>=3.1.0 in
```

```
c:\users\lenovo\anaconda3\lib\site-packages (from scikit-learn->sentence-
transformers) (3.5.0)
```

2 Import Libraries

- `WebBaseLoader` → **Loads website text.**
- `RecursiveCharacterTextSplitter` → **Splits long text into smaller chunks for embedding.**
- `SentenceTransformerEmbeddings` → **Converts chunks to vectors.**
- `Chroma` → **Stores and retrieves vectors.**
- `Ollama` → **Runs Llama3.2:1b locally.**
- `RetrievalQA` → **RAG chain combining LLM + Retriever.**

[2]:
```python
from langchain.document_loaders import WebBaseLoader
from langchain.text_splitter import RecursiveCharacterTextSplitter
from langchain.embeddings import SentenceTransformerEmbeddings
from langchain.vectorstores import Chroma
from langchain.chains import RetrievalQA
from langchain.llms import Ollama
import os
```

```
USER_AGENT environment variable not set, consider setting it to identify your
requests.
```

3 Define Trusted URLs

These are **trusted sources** for financial and business info. You can add more or replace them with any website.

[3]:
```python
urls = [
    "https://www.moneycontrol.com/stocks/marketstats/",
    "https://www.economictimes.indiatimes.com/markets",
    "https://www.financialexpress.com/market/",
    "https://www.business-standard.com/"
]
```

4 Load Website Data

- `WebBaseLoader` **scrapes text from websites.**
- `docs` **list stores all loaded documents.**
- **Errors are caught if a website fails to load.**

[4]:
```python
# Load all website content
loaders = [WebBaseLoader(url) for url in urls]
docs = []

for loader in loaders:
```

```
    try:
        data = loader.load()
        docs.extend(data)
    except Exception as e:
        print(f"Error loading {loader}: {e}")

print(f"Total documents loaded: {len(docs)}")
```

```
Total documents loaded: 4
```

5 Split Text into Chunks

- **LLMs work better with smaller chunks of text.**
- **Overlap ensures context is not lost between chunks.**

[5]:
```
splitter = RecursiveCharacterTextSplitter(
    chunk_size=1000,
    chunk_overlap=200
)
split_docs = splitter.split_documents(docs)

print(f"Total chunks created: {len(split_docs)}")
```

```
Total chunks created: 119
```

6 Create Embeddings and Vector Database

- **Converts each text chunk into a numerical vector.**
- **Stores vectors in ChromaDB for fast retrieval.**

[6]:
```
# Initialize embeddings
embedding_function = 
 ↪SentenceTransformerEmbeddings(model_name="all-MiniLM-L6-v2")

# Create vector database with Chroma
vector_store = Chroma.from_documents(split_docs, embedding_function)

print(" Vector store created successfully!")
```

```
C:\Users\Lenovo\AppData\Local\Temp\ipykernel_9708\3043873310.py:2:
LangChainDeprecationWarning: The class `HuggingFaceEmbeddings` was deprecated in
LangChain 0.2.2 and will be removed in 1.0. An updated version of the class
exists in the :class:`~langchain-huggingface package and should be used instead.
To use it run `pip install -U :class:`~langchain-huggingface` and import as
`from :class:`~langchain_huggingface import HuggingFaceEmbeddings``.
  embedding_function = SentenceTransformerEmbeddings(model_name="all-
MiniLM-L6-v2")
```

```
WARNING:tensorflow:From C:\Users\Lenovo\anaconda3\Lib\site-
packages\tf_keras\src\losses.py:2976: The name
tf.losses.sparse_softmax_cross_entropy is deprecated. Please use
tf.compat.v1.losses.sparse_softmax_cross_entropy instead.

 Vector store created successfully!
```

7 Initialize Ollama Local LLM

- **Uses Llama3.2:1b installed locally in Ollama.**
- **Fully offline; works with your RAG pipeline.**

[7]:
```python
# Initialize Ollama local LLM
llm = Ollama(model="llama3.2:1b")
```

```
C:\Users\Lenovo\AppData\Local\Temp\ipykernel_9708\461239926.py:2:
LangChainDeprecationWarning: The class `Ollama` was deprecated in LangChain
0.3.1 and will be removed in 1.0.0. An updated version of the class exists in
the :class:`~langchain-ollama package and should be used instead. To use it run
`pip install -U :class:`~langchain-ollama` and import as `from
:class:`~langchain_ollama import OllamaLLM``.
  llm = Ollama(model="llama3.2:1b")
```

8 Build RAG Chain

- **Retriever fetches top `k` relevant chunks.**
- **LLM generates an answer using retrieved context.**
- `return_source_documents=True`→ **keeps source links for citations.**

[8]:
```python
# Create retriever from vector store
retriever = vector_store.as_retriever(search_kwargs={"k": 3})

# Combine LLM + Retriever into a RAG pipeline
rag_chain = RetrievalQA.from_chain_type(
    llm=llm,
    chain_type="stuff",
    retriever=retriever,
    return_source_documents=True
)
```

9 Ask a Business Question

- **This is the core RAG query.**
- **Response includes AI-generated answer + sources.**

[10]:
```python
# Example query
```

```python
query = "What are the growth prospects and risks for renewable energy companies in India in 2025?"

# Run query through RAG
response = rag_chain.invoke({"query": query})

# Display answer
print(" Question:", query)
print("\n Answer:\n", response["result"])

# Display sources
print("\n Sources:")
for doc in response["source_documents"]:
    print("-", doc.metadata.get("source", "Unknown"))
```

```
 Question: What are the growth prospects and risks for renewable energy
companies in India in 2025?

 Answer:
 Based on the provided context, it appears that there are mixed views about the
growth prospects and risks for renewable energy companies in India in 2025.
Here's a summary of the points mentioned:

Growth Prospects:

* According to the article, AI and trade reforms could power India's next growth
wave.
* The World Bank's South Asia chief economist is also optimistic about India's
potential to become a leader in global innovation.
* Sustaining above 25,000 may unlock fresh upside for the Nifty.

Risks:

* There are some risks mentioned in the article, such as RBI intensifying
offshore market intervention to stabilize the rupee and planned provisioning
norms unlikely to hit private banks hard.
* Additionally, there is a mention of FPIs increasing bearish bets on Nifty
futures, suggesting that global investors may be skeptical about India's growth
prospects.

However, it's also worth noting that some companies in the renewable energy
sector are mentioned with positive outlooks, such as:

* WeWork India Management Ltd. sees its IPO price band at Rs 5,050 crore.
* Tata Capital's CEO shares insights on IPO pricing and growth.
* Bajaj Finance is up 2% after a strong Q2 revenue performance.

Overall, it seems that the growth prospects for renewable energy companies in
```

```
India in 2025 are promising, but there are also some risks to consider.

 Sources:
- https://www.economictimes.indiatimes.com/markets
- https://www.economictimes.indiatimes.com/markets
- https://www.economictimes.indiatimes.com/markets
```

10 Optional – Better Output Formatting

- **Makes answers readable for portfolio demos.**
- **Clearly separates summary from sources.**

[11]:
```python
def display_response(response, query):
    print("\033[1mQuestion:\033[0m", query, "\n")
    print("\033[1mAI Summary:\033[0m\n", response["result"], "\n")
    print("\033[1mSources:\033[0m")
    for doc in response["source_documents"]:
        print("-", doc.metadata.get("source", "Unknown"))

# Test improved display
query2 = "Which Indian IT companies are likely to benefit the most from AI␣
 ↪adoption in 2025?"
response2 = rag_chain.invoke({"query": query2})
display_response(response2, query2)
```

```
Question: Which Indian IT companies are likely to benefit the most from
AI adoption in 2025?

AI Summary:
 Based on the provided context, it appears that Accenture is likely to benefit
the most from AI adoption in 2025. The article mentions that Accenture's Q4
revenue rose 7% due to higher AI demand, and they are also described as "a
leading Indian IT services company." This suggests that Accenture might be a key
player in the Indian IT industry that could potentially benefit from AI adoption
in 2025.

Sources:
- https://www.economictimes.indiatimes.com/markets
- https://www.financialexpress.com/market/
- https://www.economictimes.indiatimes.com/markets
```

11 Save Vector Database

- **Saves embeddings so you don't have to rebuild every time.**
- **Makes the notebook production-ready.**

[12]:
```python
# Persist vector DB for future use
vector_store.persist()
print(" Knowledge base saved successfully!")
```

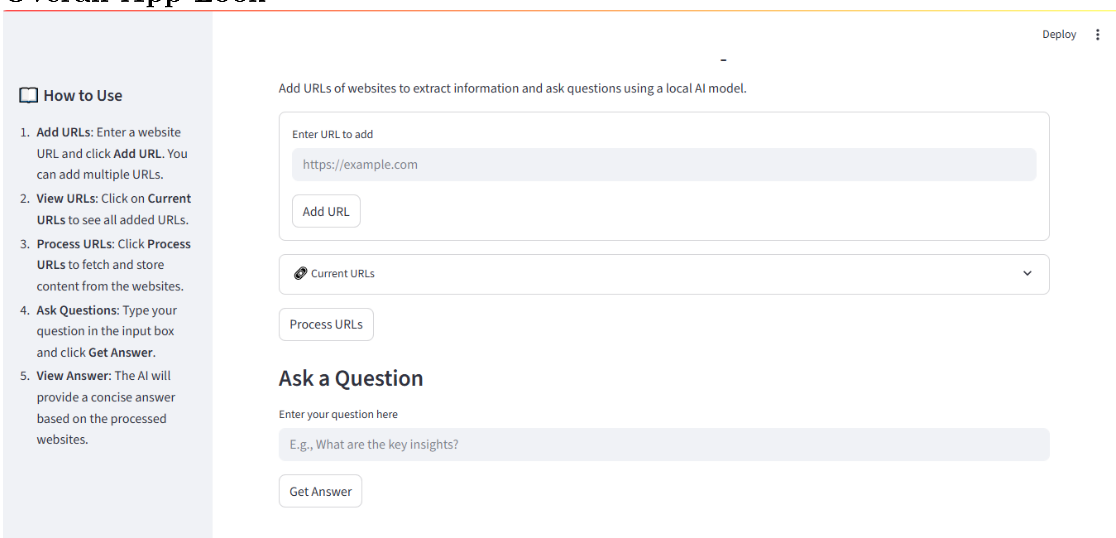
```
 Knowledge base saved successfully!

C:\Users\Lenovo\AppData\Local\Temp\ipykernel_9708\2218592760.py:2:
LangChainDeprecationWarning: Since Chroma 0.4.x the manual persistence method is
no longer supported as docs are automatically persisted.
  vector_store.persist()
```

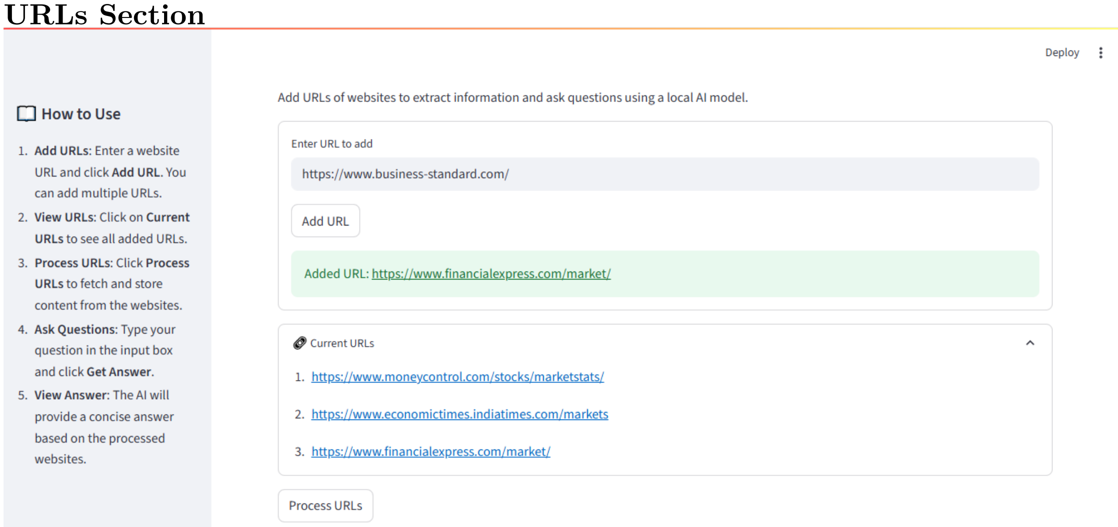
12 App Screenshots Documentation

Below are the screenshots demonstrating the functionality and usage of the AI Business Research Copilot app.

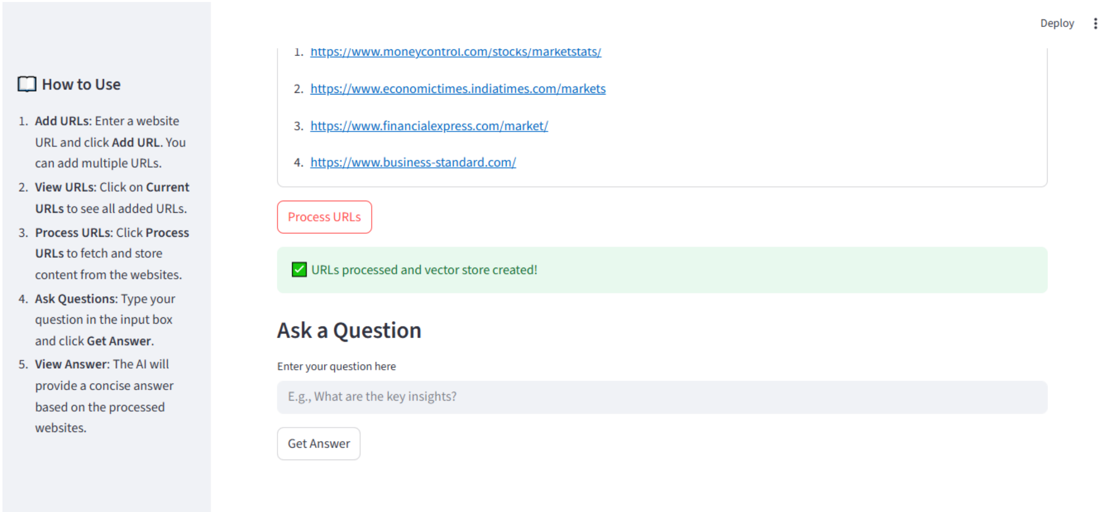
Overall App Look



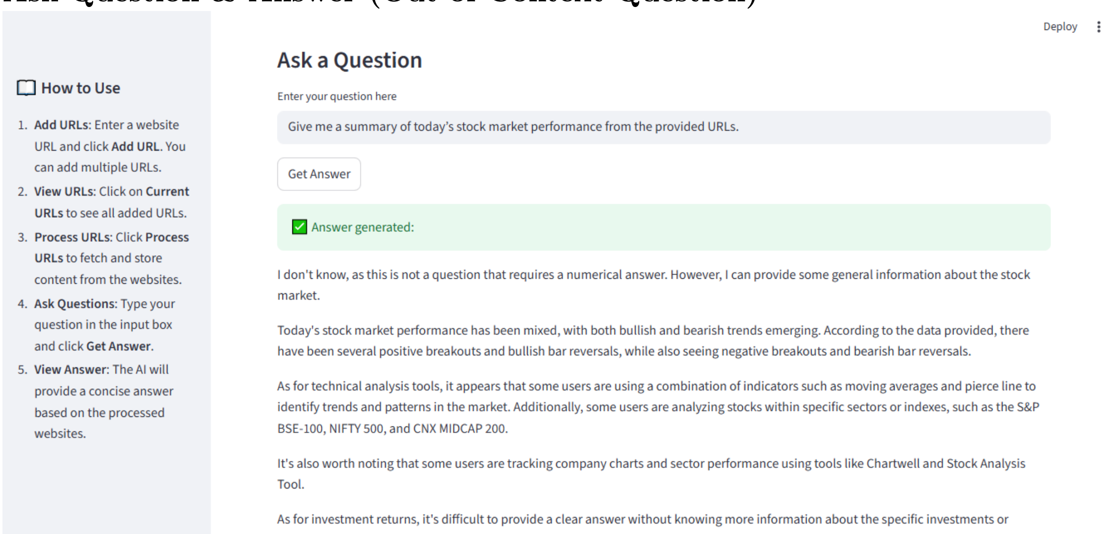
URLs Section



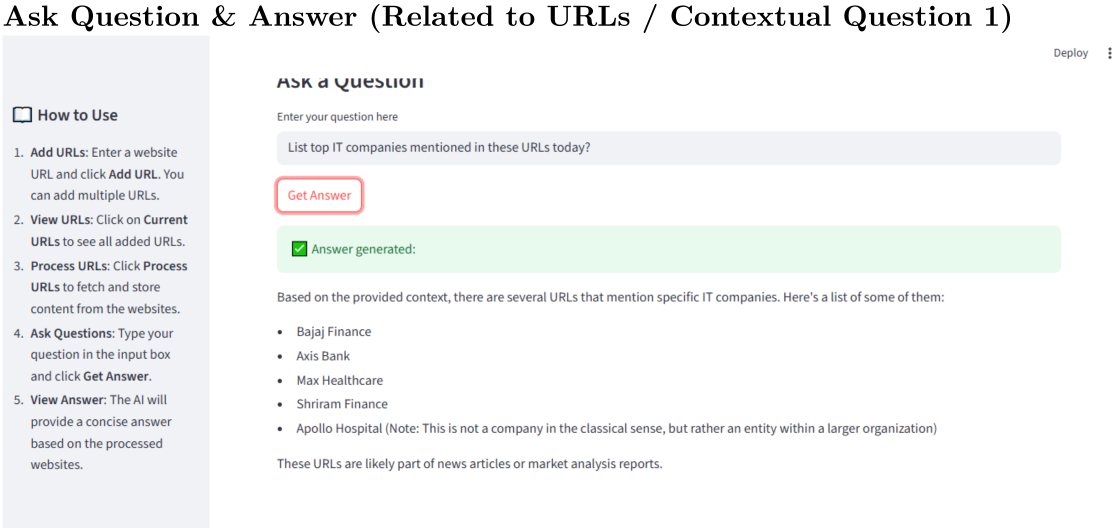
Process URLs Section



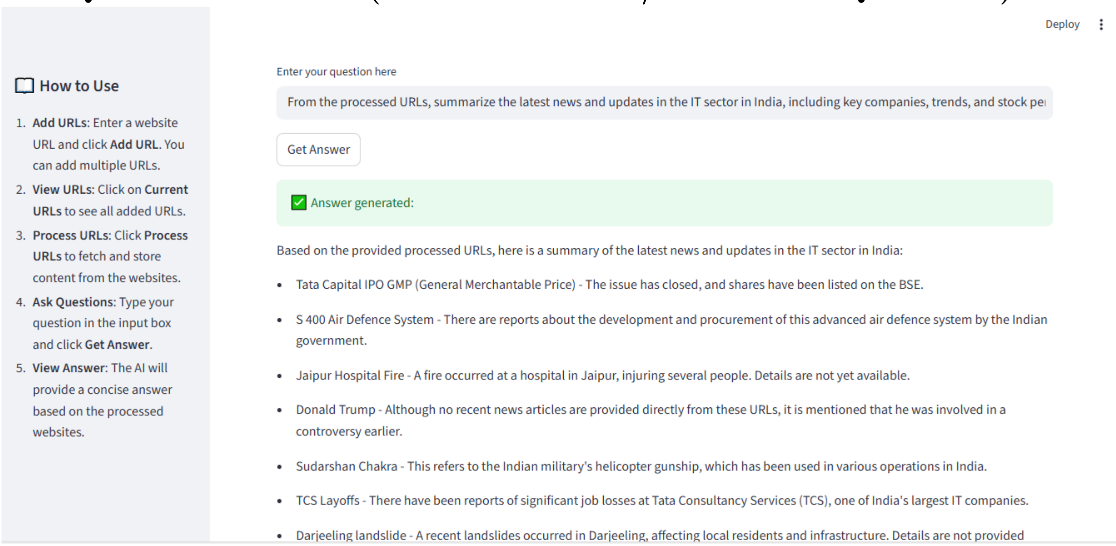
Ask Question & Answer (Out of Context Question)



Ask Question & Answer (Related to URLs / Contextual Question 1)



Ask Question & Answer (Related to URLs / Contextual Question 2)



13 Notes for Beginners

- **RAG (Retrieval-Augmented Generation):** LLM doesn't rely solely on its training; it retrieves context from documents.
- **Chunking:** Important for large websites; prevents LLM from being overloaded.
- **Embeddings:** Converts text into numbers that capture semantic meaning.
- **Vector Store:** Lets the LLM find relevant text quickly for accurate answers.
- **Ollama Local LLM:** Fully offline, secure, and fast response for sensitive data.

14 Conclusion & Summary

The **AI Business Research Copilot** is a versatile tool designed to assist users in gathering insights from multiple websites and answering questions based on the processed information.

14.1 Key Highlights:

- Users can **add multiple URLs** dynamically and process them for content extraction.
- The app **hides URLs** for a cleaner interface and allows reviewing them via a dropdown.
- Supports **contextual question answering** using a local LLM, providing intelligent responses based on the processed data.
- Designed for **ease of use**, making research faster and more efficient.

14.2 Summary:

This project demonstrates how AI can automate research tasks, extract relevant information from multiple sources, and provide meaningful answers to user queries. It can serve as a foundation for more advanced business research tools, integrating further features like trend analysis, summarization, or interactive dashboards.

> Overall, this app combines **data processing, LLM capabilities, and a user-friendly interface** into a professional research assistant.